

Machine Learning - Health Analysis and Find-S Algorithm

Question 1: Design a Machine Learning System for Health Analysis

Introduction

A Machine Learning (ML) system can predict diseases such as heart disease or diabetes using patient information like age, blood pressure, glucose level, cholesterol, BMI, and lifestyle habits. It helps doctors make faster and more accurate decisions.

Patient Data Used

- Age
- Gender
- Blood Pressure
- Glucose Level
- Cholesterol Level
- Body Mass Index (BMI)
- Smoking Habit
- Alcohol Consumption
- Physical Activity
- Family Medical History

Suitable Algorithms

- **Logistic Regression:** Used for binary classification. Easy to understand and fast.
- **Decision Tree:** Creates simple decision rules and is easy to interpret.
- **Random Forest:** High accuracy, reduces overfitting, handles missing values, and works well with medical datasets.

Steps of the ML System

1. Data Collection: Collect patient records.
2. Data Preprocessing: Remove missing values, duplicates, normalize data, and encode categorical values.
3. Feature Selection: Select important features such as age, blood pressure, glucose, BMI, and smoking status.
4. Model Training: Split data into training (80%) and testing (20%) and train the Random Forest model.
5. Model Testing: Evaluate using Accuracy, Precision, Recall, and F1-Score.
6. Disease Prediction: Predict whether the patient has heart disease or diabetes.

Architecture

Patient Data



Data Collection



Data Preprocessing



Feature Selection



Random Forest Model



Disease Prediction



Doctor's Decision

Advantages

- Early disease detection
- High prediction accuracy
- Saves time
- Supports doctors
- Improves patient care

Conclusion

Machine Learning helps predict diseases accurately. Random Forest is preferred because it provides high accuracy, reduces overfitting, and improves healthcare decision-making.

Question 2: Find-S Algorithm (Loan Approval)

Training Dataset

Employment	Salary	Credit Score	Loan History	Loan Approved
Employed	High	Good	Good	Yes
Employed	High	Good	Excellent	Yes
Self-employed	Medium	Good	Good	No
Employed	High	Excellent	Good	Yes

Steps

7. Initialize the hypothesis: $S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$.
8. After Positive Example 1: $S_1 = \langle \text{Employed}, \text{High}, \text{Good}, \text{Good} \rangle$.
9. After Positive Example 2: Generalize Loan History $\rightarrow S_2 = \langle \text{Employed}, \text{High}, \text{Good}, ? \rangle$.
10. Ignore the negative example because Find-S uses only positive examples.
11. After Positive Example 4: Generalize Credit Score $\rightarrow$ Final $S = \langle \text{Employed}, \text{High}, ?, ? \rangle$.

Final Hypothesis

$S = \langle \text{Employed}, \text{High}, ?, ? \rangle$

Conclusion

The Find-S algorithm starts with the most specific hypothesis and generalizes it using only positive examples. The final hypothesis is $\langle \text{Employed}, \text{High}, ?, ? \rangle$.